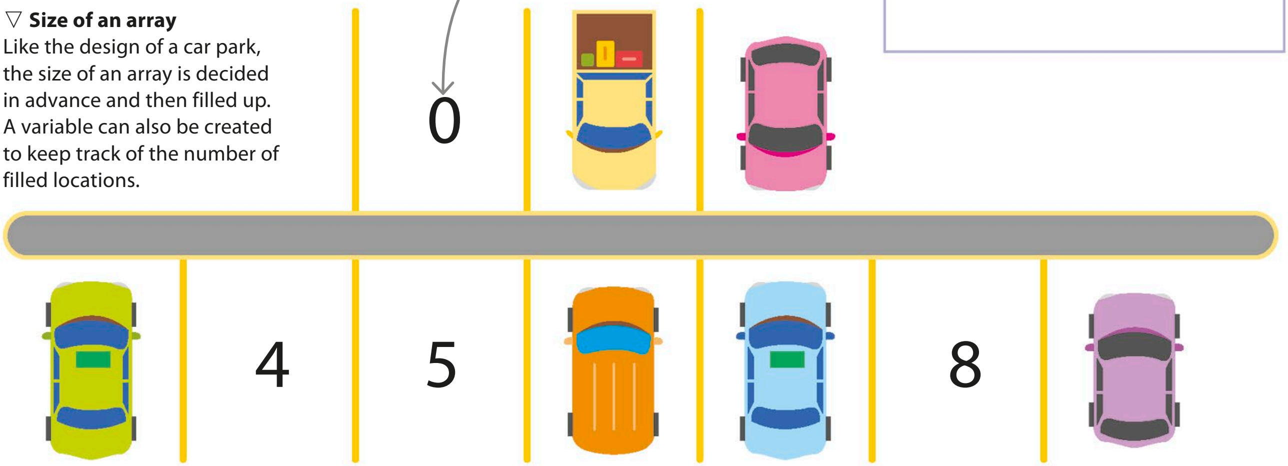
Arrays

An array is a collection of similar elements with a specific order. It is just like houses on a street or cars in a car park, which have a clear beginning and end. Location of an item in an array is called array index. The first element has a name, or label, and every other piece of data is referenced based on its distance from this first element. In an array of cars, the first would be called "car", the second "car + 1", then "car + 2", and so on.

Arrays start their numbering system with 0 instead of 1. An array with 10 elements contains locations 0–9.

▽ **Size of an array**
Like the design of a car park, the size of an array is decided in advance and then filled up. A variable can also be created to keep track of the number of filled locations.



IN DEPTH
Pros and cons

Arrays are great while accessing the data as a group, such as a music app listing all the songs of an artist. However, it can be tricky to find or modify an individual element if the array is too long. A specific piece of data could be hiding anywhere – in position 1, 53, or 5,000. You don't know until you look at each element.

Objects

An object is like a special, custom-sized variable. It is useful when data in a variety of sizes and shapes need to be grouped together. The key thing about objects is that the users have to write the code themselves. For this, a blueprint, called a class, needs to be defined. It includes listing all of the object's attributes, each of which is a different variable. Afterwards, the same blueprint can be used every time to create a new object.

Each object contains its own unique data, based on the class blueprint.

▷ **A customer object**
An object for an online shopper might contain a string of text (customer name), a couple of numbers (customer age and a unique identification number), and an array (the items in the customer's online shopping cart).

IN DEPTH
Why use objects?

The main goal of objects is to keep data in a central place. Object-oriented programming (OOP) lets you create object methods, which are special functions to manipulate object data. You can restrict which variables can be accessed by different parts of the program. This is especially useful for big software applications that have hundreds, if not thousands, of code files, each of which might process a different, unrelated type of data.